

EE2020/EE2026
Tutorial 4 - Solutions

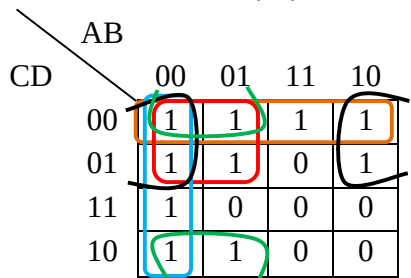
1.

Truth Table

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$Z_{SOP} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}\bar{D} + ABC\bar{D} + ABCD$$

$$Z_{POS} = (A + \bar{B} + \bar{C} + \bar{D}).(\bar{A} + B + \bar{C} + D).(\bar{A} + B + \bar{C} + \bar{D}).(\bar{A} + \bar{B} + C + \bar{D}).(\bar{A} + \bar{B} + \bar{C} + D).(\bar{A} + \bar{B} + \bar{C} + D)$$



A 4x4 Karnaugh map for variables A, B, C, and D. The horizontal axis is labeled AB with values 00, 01, 11, 10. The vertical axis is labeled CD with values 00, 01, 11, 10. The map contains 1s in the following cells: (00,00), (01,00), (11,00), (10,00), (00,01), (01,01), (10,01), (00,11), (00,10), (01,11), (01,10), (10,11), (10,10). There are five groupings: a blue horizontal group of four 1s in the top row (AB=00 to 10, CD=00); a red horizontal group of four 1s in the second row (AB=00 to 10, CD=01); a green horizontal group of four 1s in the bottom row (AB=00 to 10, CD=10); a blue vertical group of four 1s in the first column (AB=00, CD=00 to 10); and a red vertical group of four 1s in the second column (AB=01, CD=00 to 10).

$$Z_{MSOP} = \bar{A}\bar{B} + \bar{C}\bar{D} + \bar{A}\bar{C} + \bar{A}\bar{D} + \bar{B}\bar{C}$$

2.

Truth Table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$Z_{SOP} = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$Z_{POS} = (A + B + C) \cdot (A + B + \bar{C}) \cdot (A + \bar{B} + C) \cdot (\bar{A} + B + C)$$

		A	
		0	1
BC	00	0	0
	01	0	1
	11	1	1
	10	0	1

$Z_{MSOP} = AC + BC + AB$

3. (a) $Z = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D}$ with 'X' for $ABCD = 1010$

		AB			
		00	01	11	10
CD	00	0	0	0	1
	01	1	1	1	1
	11	0	0	0	0
	10	0	0	0	X

MSOP

$$Z = \bar{C}D + A\bar{B}\bar{C}$$

		AB			
		00	01	11	10
CD	00	0	0	0	1
	01	1	1	1	1
	11	0	0	0	0
	10	0	0	0	X

MPOS

$$Z = \bar{C}(A + D) \cdot (\bar{B} + D)$$

(b) $Z = (\bar{A} + B + \bar{C})(A + B + \bar{C})$ with don't cares for $ABC = 111$ and 110

		A	
		0	1
BC	00	1	1
	01	0	0
	11	1	X
	10	1	X

		A	
		0	1
BC	00	1	1
	01	0	0
	11	1	X
	10	1	X

MSOP

$$Z = B + \bar{C}$$

MPOS

$$Z = (B + \bar{C})$$

- (c) $f(x_1, \dots, x_4) = \sum m(0, 4, 5, 6, 7) + D(1, 12, 13, 14, 15)$, where D is the set of don't cares and m is the set for which $f = 1$ (this alternate shorthand notation is also used to express min terms).

		X ₁ X ₂			
		00	01	11	10
X ₃ X ₄	00	1	1	X	0
	01	X	1	X	0
	11	0	1	X	0
	10	0	1	X	0

		X ₁ X ₂			
		00	01	11	10
X ₃ X ₄	00	1	1	X	0
	01	X	1	X	0
	11	0	1	X	0
	10	0	1	X	0

MSOP

$$Z = (x_2 + \bar{x}_1 \bar{x}_3)$$

MPOS

$$Z = (x_2 + \bar{x}_3) \cdot (\bar{x}_1)$$

4.

		AB			
		00	01	11	10
CD	00	0	0	0	0
	01	0	0	1	0
	11	0	1	1	1
	10	0	0	1	0

MSOP

$$Z = ACD + BCD + ABC + ABD$$

$$Z = ACD + BCD + ABC + ABD$$

$$\bar{C} \cdot H$$

$$\bar{D} \cdot H$$

